



PRESCRIPTION SEARCH SUPPORT

Educational Web Application Radiology

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Document version

Version	Status	Date	Author	Description
1.0	Final	13/01/25	Smals	Initial version
1.1	Final	05/05/26	Smals	Include screenshots

3. Radiology

3.1. Introduction

The PSS educational web app for radiology assists prescribers in finding evidence-based imaging recommendations based on the ESR iGuide guidelines published by the European Society of Radiology (ESR).

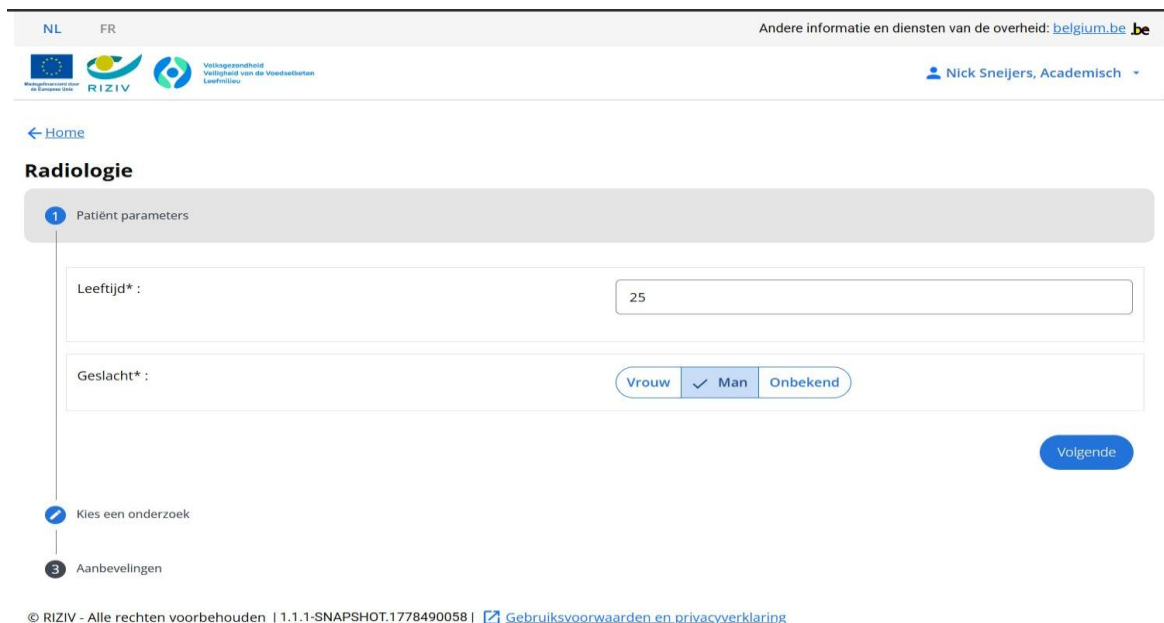
The recommendations are intended to support prescribing of imaging examinations and are not a substitute for the medical expertise of the prescriber and the radiologist.

The application consists of three sequential steps:

Step	Title	Description
1	Patient parameters	Enter the patient's age and sex
2	Choose an examination	Optionally provide an exam ID, search and select a clinical indication
3	Recommendations	Review the scored list, select an option, and provide a reason for deviation if applicable

3.2. Front-end requirements

3.2.1. Step 1: Patient Parameters



The first step collects anonymous demographic data about the patient. Both fields are mandatory before proceeding.

Field	Description
Age *	Numeric input field. Patient's age in years. Mandatory.
Sex *	Toggle buttons: Female / Male / Unknown. Mandatory. No default selection.

No personal data is stored. Age and sex are transmitted anonymously to the PSS API.

Once both mandatory fields are filled in, the Next button becomes active.

UI Labels & Translations

UI Element	NL – Nederlands	FR – Français
Age field label	Leeftijd *	Âge *

Sex field label	Geslacht *	Sexe *
Toggle – Female	Vrouw	Féminin
Toggle – Male	Man	Masculin
Toggle – Unknown	Onbekend	Non défini
Next button	Volgende	Suivant

3.2.2. Step 2: Choose an examination

NL
FR
Andere informatie en diensten van de overheid: belgium.be





Nick Snejers, Academisch

[← Home](#)

Radiologie

1 Patient parameters

2 Kies een onderzoek

Radiologie
Vul een onderzoeks-id in :

Trefwoorden voor indicaties* :

Laad de indicaties
Kies een indicatie om de richtlijnen te leren:
 Type en selecteer een indicatie*

Volgende

3 Aanbevelingen

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Step 2 collects the clinical context. There are two input fields: one optional and one mandatory.

Field 1 – Exam ID (Optional)

Allows the prescriber to specify a particular imaging examination they have in mind. When an exam is provided, the PSS backend filters the indication list to only those indications linked to AUC recommendations for that exam.

- Only one selection allowed.
- The field is entirely optional.

Field 2 – Indication Keywords (Mandatory)

The prescriber types one or more clinical keywords. Confirmed keywords appear as removable chips below the input field (e.g. "Headache x"). Multiple keywords are allowed.

After entering keywords, the prescriber clicks **Load indications / Laad de indicaties** to retrieve a filtered list of clinical indications from the PSS API (via QSI).

Indication Selection

From the returned list, the prescriber selects one or more relevant indications. Selected indications appear as removable chips (e.g. "Headache, positional x"). Multi-select is supported.

UI Labels & Translations

UI Element	NL – Nederlands	FR – Français
Exam ID field label	Vul een onderzoeks-id in	Entrez un identifiant de recherche
Keywords field label	Trefwoorden voor indicaties *	Mots clés pour les indications*
Load indications link/button	Laad de indicaties	Charger les indications
Indication selector label	Kies een indicatie om de richtlijnen te leren	Choisir une indication pour prendre connaissance des directives
Indication placeholder	Type en selecteer een indicatie*	Encodez et sélectionnez une indication*
Confirm / Next button	Volgende	Suivant

3.2.3. Step 3: Recommendations

Radiologie

- ☒ Patiënt parameters
- ☒ Kies een onderzoek
- ☒ Aanbevelingen

Relevantie	Onderzoek	Radiatie
8	MR, head, wo/w iv contrast	
7	MR, head, wo iv contrast	
7	MR, spine, incl MR myelography	
7	CT, myelography, spine	☠☠☠☠
5	CT, head, wo iv contrast	☠☠☠☠
5	CT, head, wo/w iv contrast	☠☠☠☠
4	CT, head, w iv contrast	☠☠☠☠
3	MR, angiography, head, wo iv contrast	
3	MR, angiography, head, wo/w iv contrast	
3	CT, angiography, head, w iv contrast	☠☠☠☠

[Raadpleeg de referentieliteratuur](#)

[Bevestigen](#)

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Step 3 displays the list of imaging recommendations, sorted by relevance score. The table contains three columns:

Column	Description
Relevance / Pertinence	Numeric score (1–9) shown as a proportionally filled colour bar with the number displayed inside.
Examination / Type d'imagerie	Name of the imaging examination, displayed in bold (e.g. "MR, head, wo iv contrast").
Radiation / Taux de radiation	Radiation level indicated with radiation icons (☠). MRI examinations have no icons.

Colour Coding

Each row is colour-coded based on the relevance score:

Score	Colour	Meaning
9, 8, 7	Green	Generally recommended
6, 5, 4	Yellow	Less recommended – reason for deviation required
3, 2, 1	Red	Not recommended – reason for deviation required
—	Grey	No score available

3.2.4. Step 4: Selecting a Recommendation

Every row in the table is clickable. On hover, the row is highlighted to indicate interactivity. The prescriber clicks a row to register their choice.

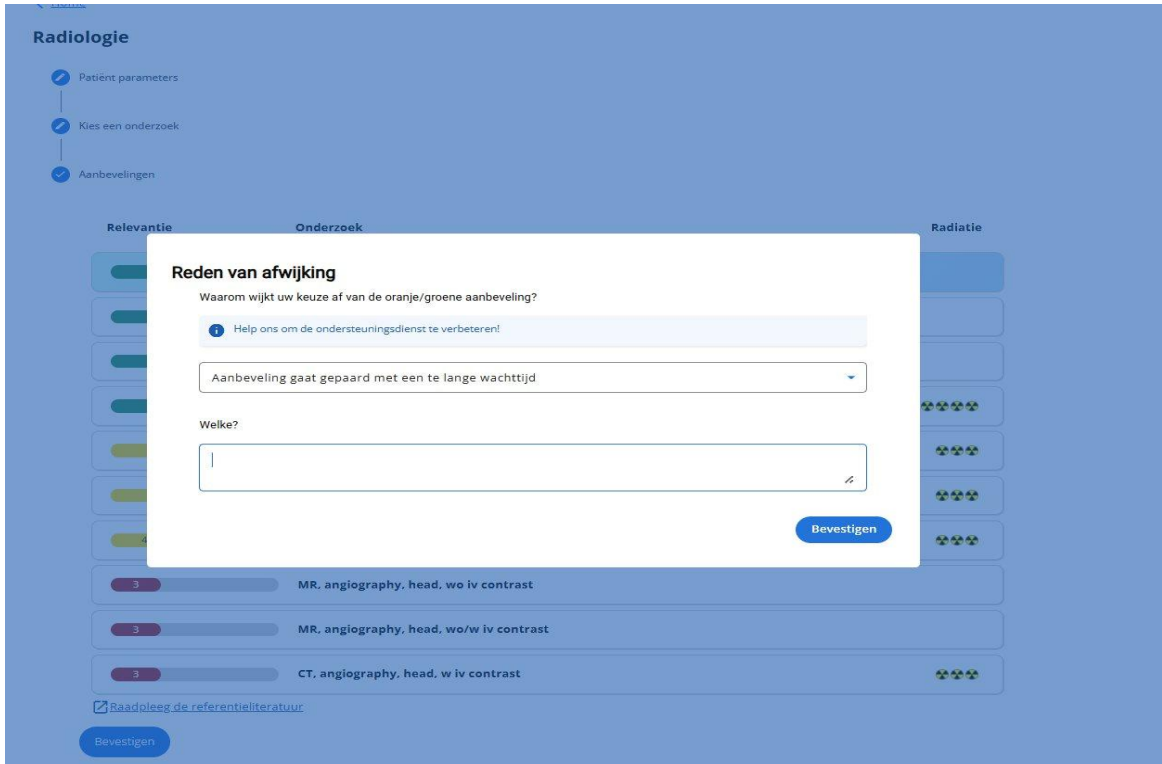
- Green score → immediate confirmation message is displayed.
- Yellow or red score → a deviation reason pop-up appears (see Section below).

Below the table, a link **Raadpleeg de referentieliteratuur / Consulter la littérature de référence** opens the full ESR iGuide source in a new browser tab.

UI Labels & Translations

UI Element	NL – Nederlands	FR – Français
Column header – Relevance	Relevantie	Pertinence
Column header – Examination	Onderzoek	Examen
Column header – Radiation	Radiatie	Radiation
Reference literature link	Raadpleeg de referentieliteratuur	Consulter la littérature de référence
Confirm button	Bevestigen	Confirmer

Reason for Deviation (Pop-up)



When the prescriber selects a recommendation with a yellow or red score, a modal window appears automatically prompting them to provide a reason for their deviation from the recommended option.

Providing a reason is mandatory for yellow and red recommendations. The Confirm button remains disabled until a reason has been selected.

Pop-up Structure

The modal contains the following elements:

- A title indicating the reason is required.
- An explanatory question asking why the prescriber deviates from the green recommendation.
- An information banner encouraging quality improvement feedback.
- A dropdown with a fixed list of 7 deviation reasons (select exactly one).
- An optional free-text field that becomes mandatory when "Other reason" is selected.
- A Confirm button to finalise the selection.

Pop-up UI Labels & Translations

UI Element	NL – Nederlands	FR – Français
Pop-up title	Reden van afwijking	Raison de la déviation
Explanatory question	Waarom wijkt uw keuze af van de oranje/groene aanbeveling?	Pourquoi votre choix diffère-t-il de la recommandation orange/verte?
Info banner	Help ons om de ondersteuningsdienst te verbeteren!	Aidez-nous à améliorer le service d'assistance!
Free-text label	Welke?	Laquelle ?
Confirm button	Bevestigen	Confirmer

Deviation Reason Dropdown – Fixed List

The dropdown contains exactly the following 7 options (exactly one must be selected):

#	NL – Nederlands	FR – Français
1	Zwangerschap	Grossesse
2	Aanwezigheid van implantaten	Présence d'implant
3	Allergie voor contrastmiddel(en)	Allergie au(x) produit(s) de contraste
4	Aanbeveling niet compatibel met de volledige klinische toestand van de patiënt	Recommandation incompatible avec l'état clinique complet du patient
5	Aanbeveling gaat gepaard met een te lange wachttijd	Recommandation s'accompagnant d'un temps d'attente trop long
6	Dit is een noodsituatie	C'est une situation d'urgence
7	Andere reden	Autre raison

4. Backend

The educational web app communicates with the PSS backend via the Rest API. The table below maps each step to its corresponding API call.

Step	FHIR Endpoint	Input	Output
1 and 2	POST /v1/domains/RADIOLOGY/indications	Patient age, sex, indication keywords, optional exam ID	List of Condition resources (clinical indications)
3	POST /v1/supportRequests/RADIOLOGY	List of Condition resources (clinical indications)	List of Recommendations
4	POST /v1/supportRequests/RADIOLOGY/{exchangeId}/conclusion	Selected Recommendation(s)	/